

GLASGOW PRESTWICK RNAS, United Kingdom

WMO: 031360

Lat: 55.5153N

Lon: 4.5847W

Elev: 89

StdP: 14.65

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.1	28.1	20.9	15.7	29.7	23.3	17.6	31.4	40.3	45.7	35.2	46.2	4.4	80	0.664

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	72.8	62.9	69.4	61.2	66.7	59.9	64.4	70.5	62.6	67.4	61.1	65.3	9.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.0	83.4	67.0	60.5	78.8	65.0	59.1	75.0	63.4	29.6	70.3	28.3	67.7	27.2	65.4	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.5	26.8	23.6	21.3	79.4	3.5	4.0	18.7	82.3	16.7	84.7	14.7	86.9	12.1	89.9
			20.6	68.3	3.6	3.3	18.0	70.6	15.9	72.6	13.9	74.4	11.3	76.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.3	40.8	41.2	43.1	46.8	51.5	56.1	59.2	59.1	55.8	50.8	45.2	41.3
	DBStd	8.29	5.30	4.99	4.64	4.64	4.84	3.95	3.69	3.64	3.94	4.78	5.44	6.58
	HDD50	1385	285	249	215	115	36	2	0	0	4	49	157	273
	HDD65	5746	749	667	678	545	418	268	185	189	278	441	594	734
	CDD50	1127	1	2	3	19	84	185	285	282	177	73	13	3
	CDD65	13	0	0	0	0	0	1	6	5	1	0	0	0
	CDH74	82	0	0	0	0	7	14	38	22	1	0	0	0
	CDH80	9	0	0	0	0	0	1	6	2	0	0	0	0
Wind	WSAvg	10.7	12.4	12.3	11.5	10.2	10.1	9.4	8.9	9.2	10.0	11.0	11.1	11.8
Precipitation	PrecAvg	37.4	3.6	2.6	2.9	2.0	2.1	2.4	2.7	3.3	4.1	4.3	4.1	3.8
	PrecMax	47.6	7.2	7.1	5.3	5.2	4.8	5.6	6.5	6.8	9.6	8.0	7.2	7.6
	PrecMin	27.8	0.5	0.2	0.7	0.2	0.3	0.8	0.6	0.7	0.8	1.4	1.0	0.9
	PrecStd	4.9	1.7	1.5	1.4	1.0	1.0	0.9	1.1	1.7	2.0	2.0	1.7	1.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	54.9	57.8	66.4	73.6	75.8	79.2	77.4	70.2	63.5	58.4	55.7
		MCWB	51.3	51.4	50.7	54.5	60.0	62.6	65.9	66.8	61.6	58.4	55.6	53.2
	2%	DB	51.5	51.9	53.6	60.0	67.6	69.8	72.6	71.1	66.6	60.9	56.1	53.7
		MCWB	49.3	49.0	49.0	51.7	57.8	60.2	63.4	63.4	60.2	56.9	53.6	51.2
	5%	DB	49.9	50.2	51.7	56.5	62.8	66.1	68.8	67.7	63.9	59.2	54.4	51.9
		MCWB	47.8	47.7	47.8	50.6	54.8	59.0	61.5	61.5	58.8	55.7	51.9	49.7
	10%	DB	48.5	48.5	50.0	54.1	59.6	63.3	65.9	65.2	61.9	57.5	52.9	50.1
		MCWB	46.4	46.1	46.7	48.9	53.2	57.1	59.9	59.9	57.3	54.3	50.4	47.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.8	51.8	52.2	55.7	62.2	64.3	67.2	68.1	63.7	59.7	56.2	53.6
		MCDB	53.4	54.1	55.6	63.2	70.6	71.7	75.5	75.1	68.3	62.2	58.0	55.3
	2%	WB	49.9	49.6	50.3	53.5	58.6	61.8	64.5	64.4	61.4	57.8	54.0	51.6
		MCDB	51.2	51.3	52.8	58.5	65.4	67.9	70.9	69.6	64.9	60.0	55.7	53.4
	5%	WB	48.3	48.0	48.6	51.2	56.1	59.5	62.5	62.5	59.7	56.3	52.4	49.9
		MCDB	49.6	49.8	50.9	55.4	61.0	64.5	67.4	66.5	62.8	58.5	54.0	51.6
	10%	WB	46.6	46.4	47.2	49.7	54.2	57.8	60.8	60.9	58.2	54.7	50.7	48.1
		MCDB	48.3	48.3	49.5	53.4	58.6	62.3	65.0	64.3	61.2	57.1	52.6	49.9
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.2	10.4	12.3	13.5	12.1	11.5	11.5	10.9	10.0	9.1	8.9
		MCDBR	8.0	9.1	11.7	16.0	19.7	17.0	16.9	15.2	13.3	10.3	8.6	8.9
		MCWBR	7.9	8.0	8.5	10.0	11.5	9.9	9.6	9.0	8.4	8.0	8.2	8.7
	5% WB	MCDBR	8.1	8.4	10.1	13.9	17.2	15.1	14.5	13.4	11.4	9.1	8.4	9.1
		MCWBR	8.2	8.0	8.2	9.5	10.8	9.7	9.0	9.0	8.4	8.0	8.4	9.1
Clear-Sky Solar Irradiance	taub	0.338	0.347	0.372	0.390	0.393	0.387	0.388	0.373	0.365	0.347	0.327	0.323	
	taud	2.202	2.298	2.272	2.261	2.303	2.353	2.357	2.412	2.430	2.404	2.290	2.183	
	Ebn at Noon	174	226	249	263	269	272	269	266	252	225	180	149	
	Edh at Noon	19	25	32	37	38	36	36	32	27	22	17	16	
All-Sky Solar Radiation	RadAvg	156	374	734	1184	1516	1491	1429	1158	806	435	209	117	
	RadStd	21	43	56	92	140	161	142	84	62	59	22	14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page